

VELOOP REWARDS

DATA ANALYST INTERNSHIP - TASK 01

FINAL CONSOLIDATED SUBMISSION

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Product Analytics, Data Readiness & Data Management Blueprint

Prepared as a student internship submission pack. All supplied and generated operational datasets are labelled simulated and are not production VELOOP statistics.

Task scope is aligned to the provided 44-page Task 01 document, including the 21 requested deliverables, data integrity rules, event design, data model, KPI catalogue, data profiling, quality controls, dashboard blueprint, developer handoff, governance and implementation backlog.

01 - Executive Summary

This project investigates what data a developing VELOOP Rewards product should collect, how the data should be structured, how data quality should be controlled, and how future analytics can be delivered reliably. The supplied workbook contains 1,000 clean simulated user records plus a 1,008-row messy audit version. Because the supplied workbook is Users-only, additional operational sheets were created strictly for analytical testing and are clearly labelled SIMULATED DATA - CREATED FOR ANALYTICAL TESTING.

- Primary finding: the biggest analytical dependency is not dashboard design; it is reliable instrumentation and stable identifiers across users, sessions, task activity, rewards, wallet transactions, referrals and withdrawals.
- The supplied Users data supports basic user/account profiling, but it does not contain the granular event and transaction history needed to calculate many product KPIs directly.
- The blueprint therefore separates Confirmed information, Proposed analytics requirements, To Be Verified implementation details, and Simulated testing data.
- The highest-priority controls are stable user/event/transaction IDs, event idempotency, controlled status values, referential integrity, timestamp standardization, and task-to-reward / transaction reversal linkage.
- The deliverable includes 34 governed KPIs, 34 event definitions, 122 data-dictionary fields, an engineer-readable ER model, data quality controls, seven dashboard blueprints, developer acceptance tests, a pipeline model, privacy roles, change management, a readiness scorecard, and a prioritized implementation backlog.

Supplied user-data observations: 1,000 records; account statuses include Active, Inactive, Suspended and Pending Verification. The source workbook documents a period of 01 Apr 2026 to 15 Sep 2026; the profiling also identified 72 missing last-login values and a small set of derived/date consistency items that should be verified before production use.

Recommendation for implementation sequence: establish the event and identity contract first, then validate data quality at collection and transformation, then publish the governed KPI layer, and only after that build production dashboards.

02 - Product Data Audit

Detailed row-level mapping is available in workbook sheets 03_Product_Data_Audit and 04_Data_Coverage_Matrix. The mapping distinguishes what the user does from what the system records and from the business outcome that management may measure.

Area	Representative feature	User action	Analytics state	Priority
Accounts	Registration/Login	Register or log in	user_registered / user_logged_in	Critical
Earning	Task-based earning	View/open/start/complete	task * lifecycle + validation	High
Rewards	Daily reward / streak / spin	Claim, maintain, spin	reward_* and engagement events	High
Wallet	Wallet ledger	Receive/consume balance	wallet_transaction_*	Critical
Withdrawals	Payout lifecycle	Request / process /	withdrawal *	Critical

Area	Representative feature	User action	Analytics state	Priority
		complete		
Referrals	Referral funnel	Create / register / activate / convert	referral_*	High
Support	Help center	View content / raise issue	help_center_viewed / support ticket	Medium

Analytics Gaps Identified

Gap	Why it matters	Priority
GAP-01 - Stable session_id not verified	Session stitching and session funnels require a stable identifier.	Critical
GAP-02 - Stable event_id + idempotency not verified	Duplicate events can overstate activity and KPI counts.	Critical
GAP-03 - Task activity-to-reward linkage not verified	Activity and reward credit must be separated and reconciled.	Critical
GAP-04 - Wallet reversal/correction linkage not verified	Ledger auditability depends on parent transaction references.	Critical
GAP-05 - Status vocabulary needs governance	Inconsistent states break KPI filters and state machines.	High
GAP-06 - Snapshot cutoff mismatch needs verification	Supplied README ends 2026-09-15 while some last_login values extend later.	High
GAP-07 - Referral conversion definition needs product approval	Registration is not the same as activation or conversion.	High
GAP-08 - Withdrawal reason categories need approved taxonomy	Supports operational analysis without inventing confidential rules.	High
GAP-09 - Versioned KPI/data dictionary ownership not established	Prevents silent metric drift.	High
GAP-10 - Automated data quality gates are not running	Bad IDs/dates/statuses can reach BI.	Critical
GAP-11 - External provider outcome fields need standardization	Third-party dependencies may cause outcomes outside VELOOP control.	Medium
GAP-12 - Support/help-center dataset is absent from supplied workbook	Support KPIs cannot be computed from Users-only data.	Medium

03-08 - Analytics Specifications

The workbook contains the detailed user, earning/reward, wallet/withdrawal, referral/engagement and event-tracking specifications. The key design decision is to keep lifecycle stages separate so that an action being performed is not automatically treated as a completed business outcome.

Domain	Core measurement	Required identifiers	Important distinction
User	Registration, login, onboarding, lifecycle	user_id, event_id, session_id	Account state vs user event
Earning	Discovery -> Start -> Completion -> Validation -> Reward	user_id, task_id, task_activity_id	Activity occurred vs reward credited
Wallet	Ledger movements and reversals	transaction_id, user_id, reference_id	Transaction vs reversal/correction
Withdrawal	Requested -> Processing -> Completed + Rejected/Reversed	withdrawal_id, user_id	Pending queue vs final outcome
Referral	Created -> Registered -> Activated -> Converted	referral_id, referring/referred user IDs	Registration is not conversion
Engagement	Feature adoption, repeat usage, daily activity	user_id, timestamp, feature	Exposure vs meaningful use

Event Tracking Standard

Naming convention: object_action (for example, task_started, reward_credited, withdrawal_completed). This keeps event names simple, predictable and easy to query. Every event specification includes a trigger, definition, identifiers, timestamp, status, required/optional properties, validation, source, business purpose, KPI dependency and implementation status. The proposed registry contains 34 meaningful events.

09-10 - Data Dictionary and Logical Data Model

The analytical model contains the core entities requested by the task: Users, Tasks, Task Activity, Rewards, Wallet Transactions, Withdrawals, Referrals and Feature Events. Support Tickets is also proposed because the task requires support/help analytics, but no support dataset was supplied.

Dataset	Records	Columns	Date range	Primary-ID check	Duplicate rows	Missing/notes
			06			
Task Activity (simulated)	6,000	15	2026-04-01 to 2026-09-17	6,000	0	-
Rewards (simulated)	5,000	13	2026-04-01 to 2026-09-17	5,000	0	-
Wallet (simulated)	3,500	13	2026-04-01 to 2026-09-17	3,500	0	-
Withdrawals (simulated)	1,200	12	2026-04-01 to 2026-09-17	1,200	0	-
Referrals (simulated)	900	10	2026-04-01 to 2026-09-17	900	0	-
Feature Events (simulated)	12,000	13	2026-04-01 to 2026-09-17	12,000	0	-
Messy Data (simulated)	501	16	2026-04-14 to 2026-09-18	479 primary IDs	1 duplicate row	824 cells

Relationship checks on generated operational sheets passed against supplied Users IDs and the generated master-task / task-activity keys. The supplied workbook itself does not contain the expected operational tables; this absence is documented as an analytics-readiness gap rather than filled in silently.

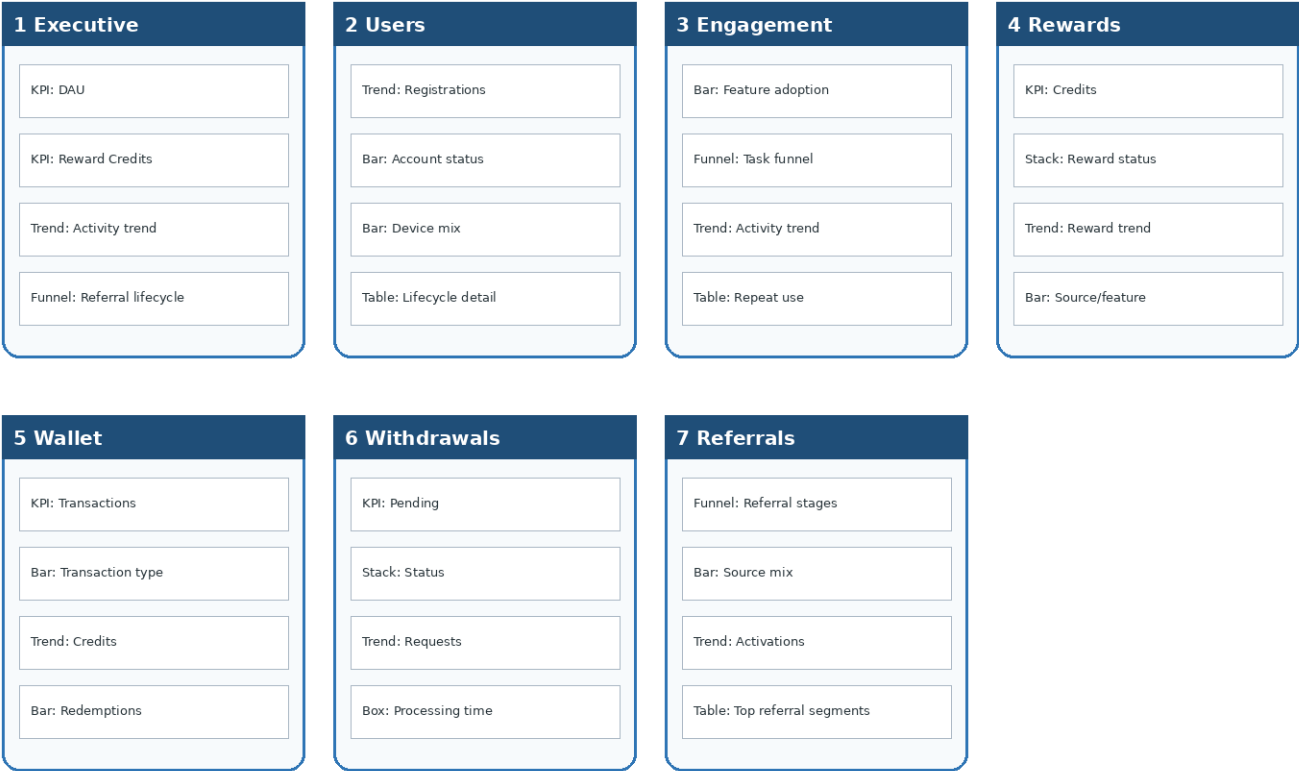
13 - Data Quality Audit and Control Framework

The supplied Users_Messy_Audit sheet contains intentional quality issues. The audit identified duplicate IDs, missing values, malformed dates and negative derived values. A second synthetic Messy Data sheet was created to test invalid status, whitespace, negative numeric and orphan-reference controls.

Issue	Dataset	Records / rows	Severity	Control
Duplicate IDs	Users_Messy_Audit	8	High	Unique-key validation + quarantine
Malformed registration date	Users_Messy_Audit	2	High	Datetime schema validation
Missing last login	Users_Messy_Audit	73	Low/Medium	Conditional completeness rule
Missing age/city/source/device/status fields	Users_Messy_Audit	1 each	Medium	Required/conditional field checks
Negative days_since_registration	Users_Messy_Audit	4	Medium	Snapshot/cutoff validation
Missing/invalid IDs	Messy_Data_SIM	21 missing IDs + injected issues	High	ID validation before KPI use
Invalid status / whitespace	Messy_Data_SIM	25	Medium	Enum + trim normalization
Negative / invalid reward amount	Messy_Data_SIM	21	High	Numeric range validation
Orphan user references	Messy_Data_SIM	8	High	Referential integrity check

Control layers: Application -> Data Collection -> Raw Data -> Transformation -> Analytics Layer -> Dashboard. Critical controls include ID validation, event schema validation, duplicate transaction/event detection, timestamp standardization, referential integrity, lifecycle-state checks, unit-aware aggregation, metric-version mapping and BI freshness checks.

14 - Dashboard Blueprint



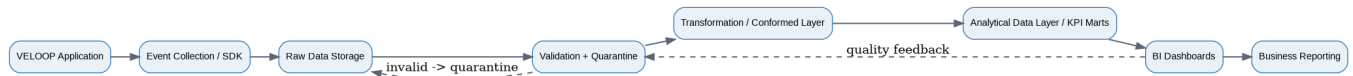
Seven future dashboards are specified: Executive Overview, User Analytics, Engagement, Rewards, Wallet, Withdrawals and Referrals. Each visual is defined in the workbook with metric, calculation, source, visualization, filters, dimensions, date range, refresh frequency, business question and decision supported.

15 - Developer Analytics Handoff

Critical event specifications are written so a developer can implement them without guessing. Each requirement contains a requirement ID, event, trigger, payload, validation, deduplication approach, timestamp rule, source, KPI dependency and acceptance test. High-priority examples include `task_completed`, `reward_credited`, `wallet_transaction_created`, `withdrawal_requested` and `withdrawal_completed`.

Requirement	Event	Key acceptance test	Priority
REQ-05	<code>task_completed</code>	Exactly one analytically usable completion event per qualifying task instance; replay does not double-count.	High
REQ-07	<code>reward_credited</code>	Reward credit references the source activity/transaction where applicable and records amount/unit/status.	High
REQ-08	<code>wallet_transaction_created</code>	Unique <code>transaction_id</code> is emitted and accepted once.	High
REQ-10	<code>withdrawal_requested</code>	Exactly one request record exists for each accepted request.	High
REQ-11	<code>withdrawal_completed</code>	Completion event includes final timestamp/status and references the <code>withdrawal_id</code> .	High
REQ-12	<code>referral_converted</code>	Conversion event uses the approved business definition and <code>referral_id</code> .	Medium

16-17 - Data Pipeline, Operating Model, Privacy & Governance



Role	Access model	Key rule
Engineering	Implementation-level analytics events	No passwords, API keys, auth secrets or payment credentials.
Data Analyst	Governed analytical datasets	Use approved/sanitized fields and stable IDs.
Product/Ops	Aggregated business reporting	Consume governed KPIs rather than raw credentials.
Content	Aggregated content performance	Use content IDs/categories and approved aggregates.
Intern	Approved/sanitized datasets only	Use simulated data or explicitly approved sanitized extracts.

The proposed architecture intentionally uses a practical layered flow rather than unnecessary infrastructure: application -> events -> raw storage -> quality gate -> transformation -> analytical layer -> BI -> reporting. Monitoring should cover freshness, failed loads, rejected records and KPI/data-quality changes.

18-20 - Change Management, Readiness, Assumptions

Analytics change management uses a versioned Metric/Event Change Log. Any feature, event, reward mechanic, withdrawal method, field, KPI or dashboard change must record the old and new definition, reason, affected KPIs/dashboards/reports, owner and version.

Readiness area	Score	Evidence	Next action
Product data mapping	4/5	22 product/features mapped	Confirm release-level implementation states
Event coverage	2/5	34 events specified; most proposed	Implement critical events
Logical model	4/5	Core entities + relationships documented	Engineering review
Data dictionary	5/5	122 fields documented	Assign operational owners
KPI definitions	4/5	34 reproducible definitions	Stakeholder approval
Data quality	4/5	Supplied + synthetic quality tests	Automate gates
Dashboard readiness	3/5	7 dashboard blueprint	Build after governed KPI layer
Governance	4/5	Role-based access matrix	Approve final permissions

Mandatory assumptions are recorded in the workbook with owner, risk and verification status. Major unknowns are stable event_id/session_id availability, final withdrawal state rules, referral conversion definition, reward reversal linkage, snapshot cutoff and provider outcome fields.

21 - Final Implementation Backlog

Priority	Requirement	Reason	Stage
Critical	Stable user_id across events/transactions	Required for cross-domain joins and cohorting.	Pre-launch
Critical	Stable event_id + idempotency	Required to prevent duplicate analytics.	Pre-launch
Critical	Task activity -> reward linkage	Separates activity from actual reward credit.	Pre-launch
Critical	Transaction ID + reversal linkage	Required for wallet auditability.	Pre-launch
Critical	Controlled event/status taxonomy	Prevents definition drift.	Pre-launch
Critical	Referential integrity controls	Prevents orphan records.	Pre-launch
High	Withdrawal lifecycle instrumentation	Enables queue and processing analysis.	Pre-launch
High	Referral lifecycle instrumentation	Separates registration, activation and conversion.	Pre-launch
High	Timestamp standard/timezone	Required for accurate day-level reporting.	Pre-launch
High	Versioned KPI/data dictionary	Protects dashboards from silent metric changes.	Pre-launch
High	Automated data-quality gates	Stops invalid records from reaching BI.	Pre-launch
Medium	Governed BI dashboard layer	Enables recurring management reporting after production data exists.	Post-launch

Priority	Requirement	Reason	Stage
Medium	Support/help-center instrumentation	Enables support/self-service KPIs.	Post-launch
Medium	Advanced segmentation	Useful after core instrumentation is stable.	Post-launch
Low	Experimentation metadata	Longer-term product measurement enhancement.	Later

Final Conclusion

The project delivers an analytics foundation rather than a fabricated performance story. The supplied dataset is treated as simulated; additional operational data is explicitly labelled as simulated testing data. The most valuable next step is to move the Critical identity, event, transaction, state and quality controls into the product/data implementation, then replace synthetic testing tables with governed production data. This keeps future dashboards and KPIs traceable, reproducible and safer to operate.

References and Source Register

- Provided VELOOP Rewards Data Analyst Internship - Task 01 PDF (44 pages).
- Provided VElLoop_Rewards_Data_01.xlsx (Users_1000_Clean and Users_Messy_Audit).
- Official VELOOP Rewards Help Center: <https://www.velooprewards.in/help-center> (reviewed 2026-10-01; used for public feature terminology only, not as proof of internal implementation).
- Master workbook: VELOOP_Task01_Master_Analytics_Workbook.xlsx.